

Hypothesis: Less than $\frac{1}{2}$ of UVA full-time undergraduate students believe a hot dog is a sandwich

Pre-analysis Steps

We collected the raw data from the Google Sheets by downloading it as an Excel file. We removed three responses, who were not full-time undergraduate UVA students. We were left with responses from only full-time undergraduate UVA students. The other columns, "Year" and "Is a Hotdog a Sandwich?", did not need any further changes because they were fixed-answer questions. So we were left with 76 of the 79 original responses after cleaning. We then read the Excel file into a pandas dataframe.

Analysis Methods

For our analysis of the cleaned data, we created exploratory charts to demonstrate the total number of yes and no responses to the question of whether a hot dog is a sandwich. We also looked further into the number of yes or no responses based on the students' class year. Lastly, we looked at the percentage of yes responses by class year.

Evaluation of Success

In order to evaluate the success of our analysis, we will look to see the percentage of undergraduate students who agree a hotdog is a sandwich, and those who do not agree a hotdog is a sandwich. We can not say the exploratory analysis is successful because we are not able to determine the general opinion of UVA students. We did not get many first-year students' opinions on the matter, and we did get mostly 4th-year students' opinions, as shown on the graph. Since the distribution of the data is skewed towards upperclassmen and not representative of UVA's true population, we cannot come to a conclusion on our hypothesis.

Distribution:

1. Year Representation:

1st: 0 students (0.0%)
2nd: 11 students (14.5%)
3rd: 10 students (13.2%)
4th: 55 students (72.4%)

2. Opinion by Year:

2nd: 9.1% say Yes
3rd: 20.0% say Yes
4th: 27.3% say Yes

Out of 76 UVA students:

18 say YES (23.7%) - hot dog IS a sandwich
58 say NO (76.3%) - hot dog IS NOT a sandwich

